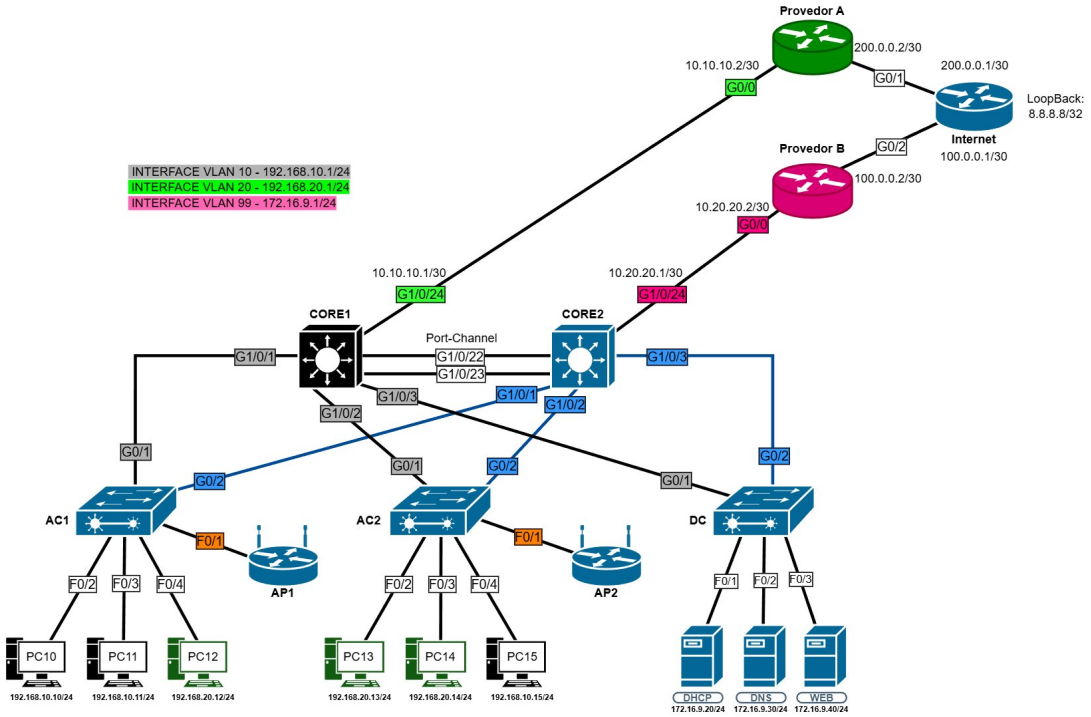
Descricao do Lab: Rede hierarquica completa com dois switches Core (CORE1 e CORE2), tres switches de acesso (AC1, AC2, DC), VLANs segmentadas, EtherChannel LACP, roteamento estatico, NAT, DHCP, DNS e servidor WEB. Dois provedores simulam redundancia de link WAN.

Dispositivo	Funcao	Enderecos Principais
CORE1	Switch L3 principal	VLAN10:.10.1 | VLAN20:.20.1 | VLAN99:.9.1 | VLAN666:.66.1
CORE2	Switch L3 redundante	G1/0/24: 10.20.20.1/30
AC1 / AC2	Switches de acesso	PCs VLAN10 e VLAN20
DC	Datacenter / servidores	DHCP:.9.20 | DNS:.9.30 | WEB:.9.40
Provedor A	ISP primario (NAT)	200.0.0.1/30 <-> 200.0.0.2/30
Internet	Loopback simulado	8.8.8.8/32

1. Topologia da Rede



Topologia completa — CORE1, CORE2, AC1, AC2, DC | Provedor A (NAT) + Provedor B (redundancia)

2. VLANs Configuradas (show vlan brief)

VLANs criadas e ativas no CORE1

APROVADO

```
CORE1# show vlan brief
```

✔ *VLAN 10, 20, 99 e 666 — status active*

Portas de acesso associadas corretamente

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✔ *Ports listadas por VLAN sem sobreposicao*

```
VLAN Name                             Status    Ports
---- -------------------------------- ---------
------------------------------
1    default                          active    Gig1/0/4, Gig1/0/5,
Gig1/0/6, Gig1/0/7
                                                Gig1/0/8, Gig1/0/9,
Gig1/0/10, Gig1/0/11
                                                Gig1/0/12, Gig1/0/13,
Gig1/0/14, Gig1/0/15
                                                Gig1/0/16, Gig1/0/17,
Gig1/0/18, Gig1/0/19
                                                Gig1/0/20, Gig1/0/21,
Gig1/1/1, Gig1/1/2
                                                Gig1/1/3, Gig1/1/4
10   VLAN0010                         active
20   VLAN0020                         active
99   VLAN0099                         active
666  VLAN0666                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

show vlan brief — VLANs 10, 20, 99 e 666 ativas no CORE1

3. Interfaces Trunk (show interfaces trunk)

Po10 em modo trunking com encapsulamento 802.1q — APROVADO

```
CORE1# show interfaces trunk
```

✔ *Po10 auto / n-802.1q / trunking / Native VLAN 1*

Gig1/0/1, 1/0/2 e 1/0/3 em trunk com 802.1q — APROVADO

✔ *Tres interfaces uplink em modo on / trunking*

VLANs 1,10,20,99,666 ativas e no STP forwarding — APROVADO

✔ *Allowed and active + Spanning tree not pruned*

```
CORE1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po10        auto         n-802.1q       trunking      1
Gig1/0/1    on           802.1q         trunking      1
Gig1/0/2    on           802.1q         trunking      1
Gig1/0/3    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po10        1-1005
Gig1/0/1    1-1005
Gig1/0/2    1-1005
Gig1/0/3    1-1005

Port        Vlans allowed and active in management domain
Po10        1,10,20,99,666
Gig1/0/1    1,10,20,99,666
Gig1/0/2    1,10,20,99,666
Gig1/0/3    1,10,20,99,666

Port        Vlans in spanning tree forwarding state and not pruned
Po10        1,10,20,99,666
Gig1/0/1    1,10,20,99,666
Gig1/0/2    1,10,20,99,666
Gig1/0/3    1,10,20,99,666
```

show interfaces trunk — Po10 + Gig1/0/1-3 em trunking com todas as VLANs

4. EtherChannel LACP (show etherchannel summary)

Grupo 10 — Port-channel Po10 ativo com protocolo LACP — APROVADO

```
CORE1# show etherchannel summary
```

✔ *Po10(SU) — Gig1/0/22(P) + Gig1/0/23(P) — SU = layer2 Up*

```
Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------
+-----------------------------------------------

10     Po10(SU)           LACP   Gig1/0/22(P) Gig1/0/23(P)
```

show etherchannel summary — Po10 LACP ativo com G1/0/22 e G1/0/23 bundled

5. Interfaces do CORE1 (show ip interface brief)

- **G1/0/24 com IP 10.10.10.1 — link para Provedor A** — APROVADO
  `CORE1# show ip interface brief`
  ✔ *GigabitEthernet1/0/24 — 10.10.10.1 — up/up*
- **VLAN10 = 192.168.10.2 | VLAN20 = 192.168.20.2** — APROVADO
  ✔ *SVIs das VLANs de usuario — up/up*
- **VLAN99 = 172.16.9.2 | VLAN666 = 192.168.66.2** — APROVADO
  ✔ *SVI de gerencia e VLAN de servidores — up/up*
- **Po10 (Port-channel) — up/up sem IP (L2)** — APROVADO
  ✔ *EtherChannel operacional como interface de trunk*

```
CORE1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Port-channel10         unassigned      YES unset  up                    up
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  down                  down
GigabitEthernet1/0/5   unassigned      YES unset  down                  down
GigabitEthernet1/0/6   unassigned      YES unset  down                  down
GigabitEthernet1/0/7   unassigned      YES unset  down                  down
GigabitEthernet1/0/8   unassigned      YES unset  down                  down
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  up                    up
GigabitEthernet1/0/23  unassigned      YES unset  up                    up
GigabitEthernet1/0/24  10.10.10.1      YES manual up                    up
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.2    YES manual up                    up
Vlan20                 192.168.20.2    YES manual up                    up
Vlan99                 172.16.9.2      YES manual up                    up
Vlan666                192.168.66.2    YES manual up                    up
```

show ip interface brief — todas as SVIs e interfaces relevantes up/up

6. Tabela de Roteamento (show ip route)

Gateway of last resort definido — 200.0.0.1 — APROVADO

```
PROVEDOR-A# show ip route
```

✔ *S* 0.0.0.0/0 via 200.0.0.1 — rota default configurada*

Rota estatica para 172.16.9.0/24 via 10.10.10.1 — APROVADO

✔ *S 172.16.9.0 [1/0] via 10.10.10.1*

Rotas estaticas para redes internas 192.168.x.0/24 — APROVADO

✔ *S 192.168.10/20/66.0 via 10.10.10.1*

Links diretamente conectados — C e L presentes — APROVADO

✔ *10.10.10.0/30 e 200.0.0.0/30 — Connected + Local*

```
Gateway of last resort is 200.0.0.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0
L       10.10.10.2/32 is directly connected, GigabitEthernet0/0
     172.16.0.0/24 is subnetted, 1 subnets
S       172.16.9.0/24 [1/0] via 10.10.10.1
S    192.168.10.0/24 [1/0] via 10.10.10.1
S    192.168.20.0/24 [1/0] via 10.10.10.1
S    192.168.66.0/24 [1/0] via 10.10.10.1
     200.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.0.0.0/30 is directly connected, GigabitEthernet0/1
L       200.0.0.2/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 200.0.0.1
```

show ip route no Provedor A — rotas estaticas + default route + connected

7. NAT — Traducao de Enderecos (show ip nat translations)

Traducoes ICMP ativas — PC interno acessando 8.8.8.8 — APROVADO

```
PROVEDOR-A# show ip nat translations
```

✔ *Inside local 192.168.10.101 -> Inside global 200.0.0.2*

4 entradas icmp — 4 pacotes do ping traduzidos — APROVADO

✔ *IDs :1 a :4 confirmam traducao NAT funcionando*

```
PROVEDOR-A#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside
global
icmp 200.0.0.2:1       192.168.10.101:1   8.8.8.8:1
8.8.8.8:1
icmp 200.0.0.2:2       192.168.10.101:2   8.8.8.8:2
8.8.8.8:2
icmp 200.0.0.2:3       192.168.10.101:3   8.8.8.8:3
8.8.8.8:3
icmp 200.0.0.2:4       192.168.10.101:4   8.8.8.8:4
8.8.8.8:4
```

show ip nat translations — 192.168.10.101 traduzido para 200.0.0.2 ao acessar 8.8.8.8

8. DHCP — Pools e Concessao de IP

Pools DHCP para VLAN 10, VLAN 20 e VLAN 666 — APROVADO

```
Servidor DHCP — IP Pools configurados
```

✔ *Gateways, DNS e ranges corretos para cada VLAN*

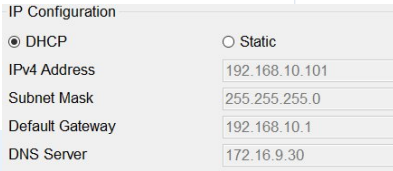
PC10 recebeu IP via DHCP — 192.168.10.101 — APROVADO

```
PC10 > Desktop > IP Configuration > DHCP
```

✔ *IP .10.101 | GW .10.1 | DNS 172.16.9.30 — automatico*

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VLAN 666	192.168.66.1	172.16.9.30	192.168.66.100	255.255.255.0	156	0.0.0.0	0.0.0.0
VLAN 20	192.168.20.1	172.16.9.30	192.168.20.100	255.255.255.0	156	0.0.0.0	0.0.0.0
VLAN 10	192.168.10.1	172.16.9.30	192.168.10.100	255.255.255.0	156	0.0.0.0	0.0.0.0

Pools DHCP — VLAN10 / 20 / 666 configuradas no servidor



PC10 com IP 192.168.10.101 c

9. Testes de Conectividade (ping)

Ping para 8.8.8.8 (Internet) — 4/4 pacotes — 0% loss — APROVADO

```
C:\> ping 8.8.8.8
```

✔ *Reply from 8.8.8.8 — TTL=253 — NAT + roteamento OK*

Ping inter-VLAN — PC10 (VLAN10) -> PC13 (VLAN20) — APROVADO

```
C:\> ping 192.168.20.13
```

✔ *3/4 — 1 timeout no primeiro pacote (ARP) — normal*

Ping para servidor DNS 172.16.9.30 — 4/4 — 0% loss — APROVADO

```
C:\> ping 172.16.9.30
```

✔ *Reply from 172.16.9.30 — TTL=127 — roteamento OK*

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.13

Pinging 192.168.20.13 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.13: bytes=32 time<1ms TTL=127
Reply from 192.168.20.13: bytes=32 time<1ms TTL=127
Reply from 192.168.20.13: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.13:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.16.9.30

Pinging 172.16.9.30 with 32 bytes of data:

Reply from 172.16.9.30: bytes=32 time<1ms TTL=127
Reply from 172.16.9.30: bytes=32 time<1ms TTL=127
Reply from 172.16.9.30: bytes=32 time<1ms TTL=127
Reply from 172.16.9.30: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.9.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Pings: 8.8.8.8 (internet) | 192.168.20.13 (inter-VLAN) | 172.16.9.30 (DNS servidor)

10. Servicos — DNS e HTTP

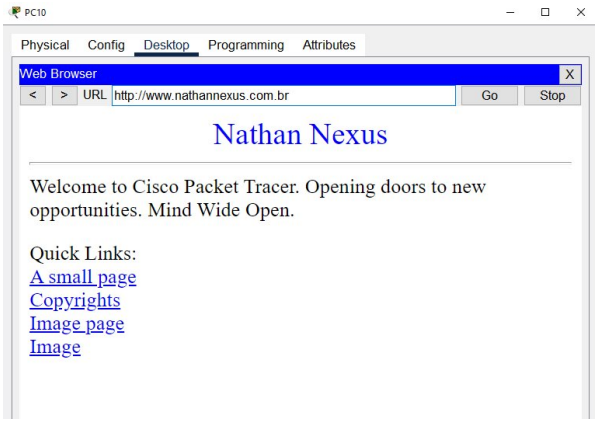
DNS resolveu www.nathannexus.com.br para o servidor WEB — APROVADO

```
PC10 > Web Browser > http://www.nathannexus.com.br
```

✔ *Pagina 'Nathan Nexus' carregada com sucesso*

Servidor HTTP respondeu na porta 80
✔ *Conteudo HTML renderizado no browser do Packet Tracer*

APROVADO



PC10 acessando www.nathannexus.com.br — DNS + HTTP funcionando

Resumo dos Resultados

Teste	Resultado	Status
VLANs (10, 20, 99, 666)	Todas ativas com portas corretas	**APROVADO**
Trunk (Po10 + G1/0/1-3)	802.1q — VLANs permitidas e ativas	**APROVADO**
EtherChannel LACP	Po10(SU) — G1/0/22+23 bundled	**APROVADO**
IP Interface Brief	SVIs + uplinks up/up	**APROVADO**
Roteamento (estatico)	Default + rotas internas configuradas	**APROVADO**
NAT (Provedor A)	192.168.10.101 -> 200.0.0.2	**APROVADO**
DHCP	PC10 obteve IP .10.101 automaticamente	**APROVADO**
Ping Internet (8.8.8.8)	4/4 — 0% loss — TTL=253	**APROVADO**
Ping inter-VLAN	3/4 — 1 ARP timeout (normal)	**APROVADO**
DNS + HTTP	nathannexus.com.br carregado	**APROVADO**

Conclusao: Todos os 10 testes foram aprovados. O laboratorio demonstra com sucesso uma rede hierarquica completa com segmentacao por VLANs, redundancia via EtherChannel LACP, roteamento estatico inter-VLAN, acesso a internet via NAT, distribuicao de IPs por DHCP e resolucao de nomes por DNS com acesso HTTP.